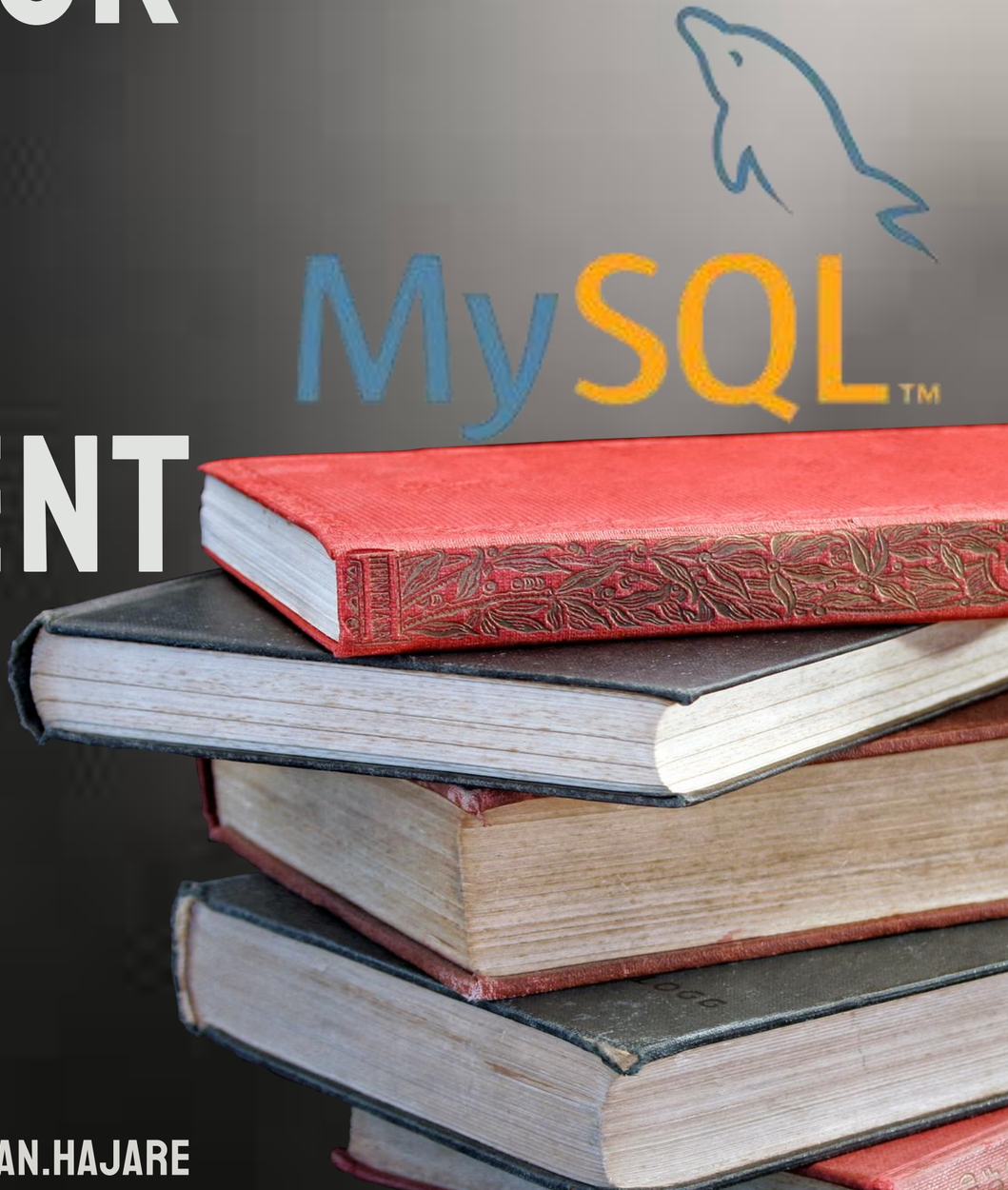
ONLINE BOOK STORE MANAGEMENT SYSTEM





PRESENTED BY -KRUSHNAVARDHAN.HAJARE

ONLINE BOOK STORE MANAGEMEBT SYSTEM

“In this project, there are three main tables: Books, Customers, and Orders. The Books table stores book details, the Customers table stores customer information, and the Orders table keeps track of all purchase transactions.”

PROJECT AIM

““The aim of this project is to create an efficient system for managing book store operations and to perform data analysis for better decision making.



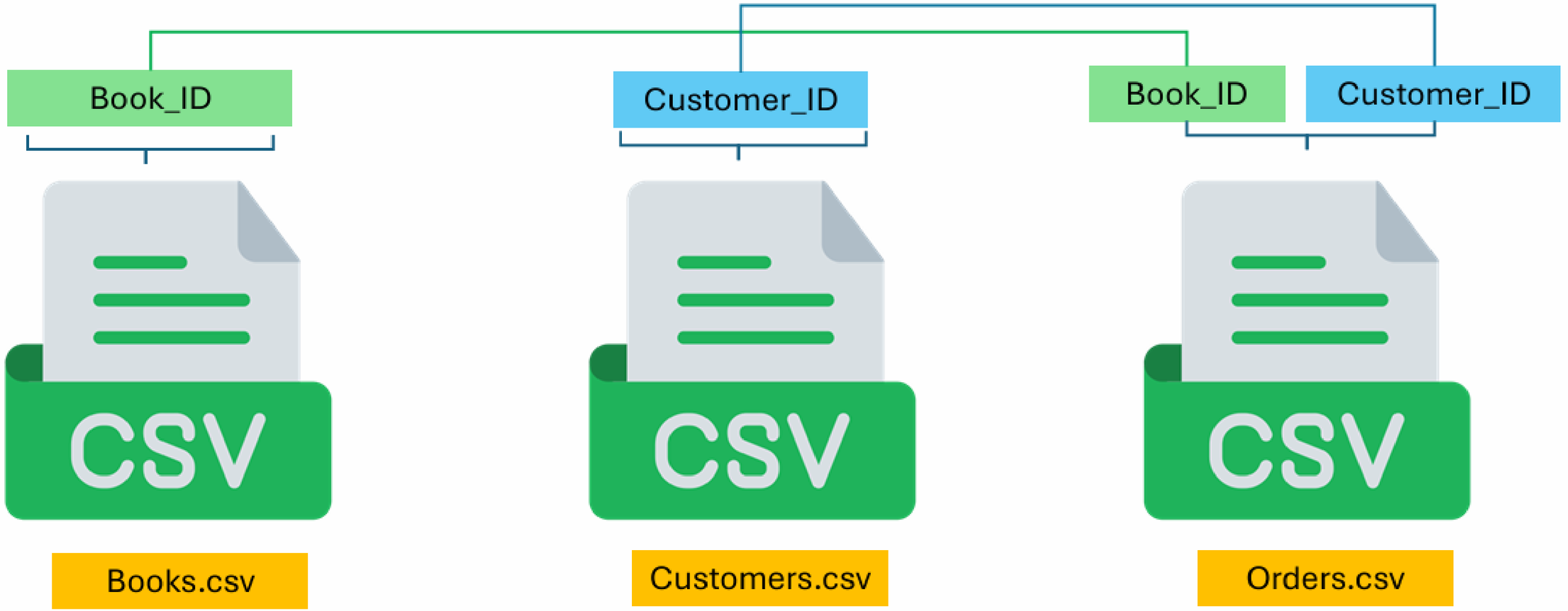
PROJECT OBJECTIVE

- To design a database for an online book store
- To store data of books, customers, and orders
- To perform CRUD operations (Create, Read, Update, Delete)
- To analyze sales and customer behavior using SQL queries
- To improve data management and decision making
- To ensure data accuracy and consistency
- To manage inventory (stock) efficiently



3 CSV Files

Tables must have at least one common column with same column name and same data type

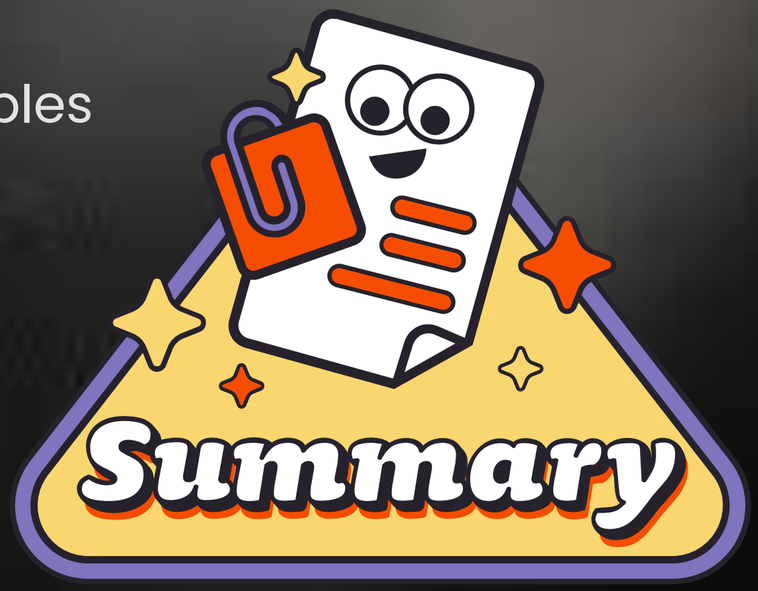


BASIC SQL QUERY SUMMARY

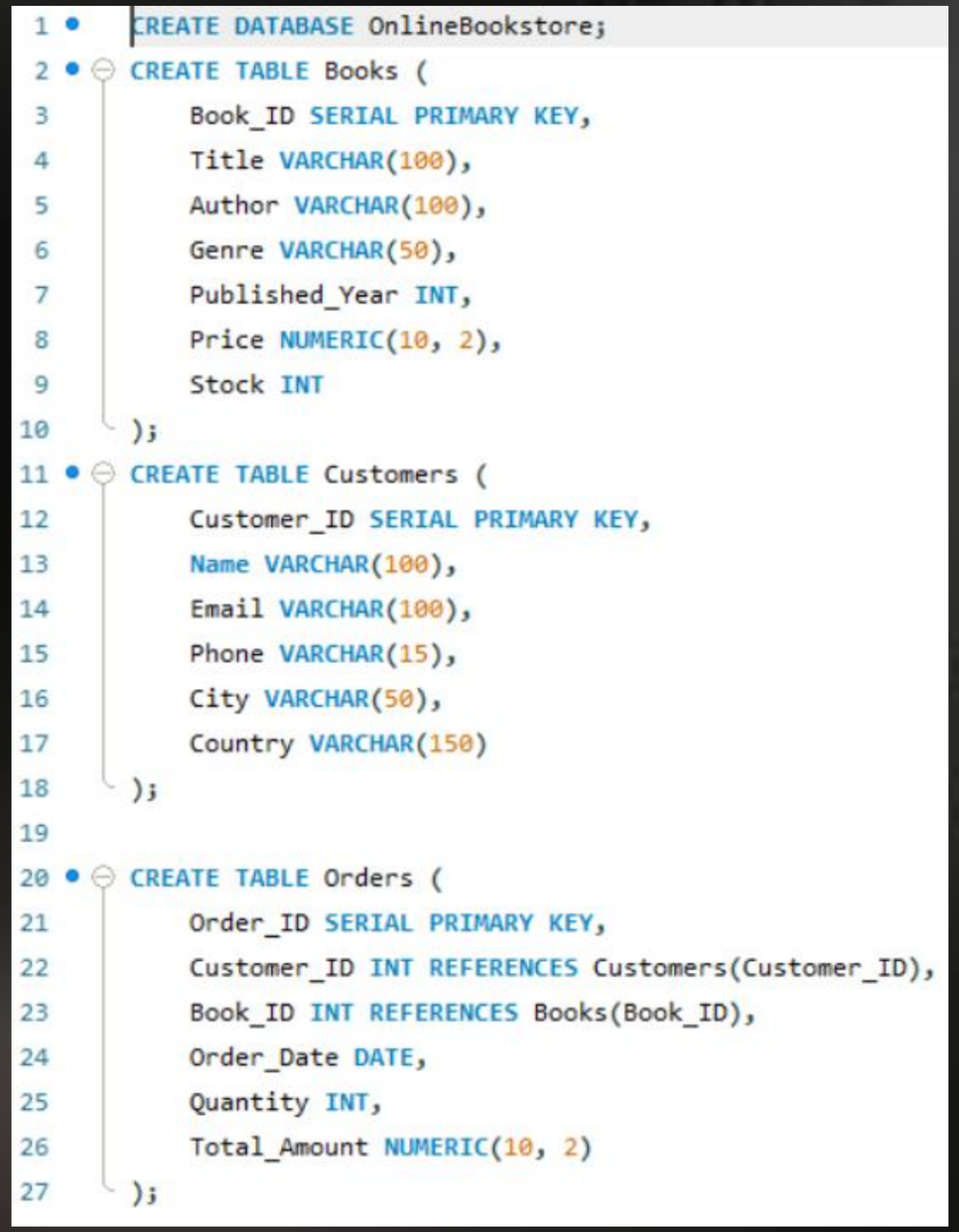
- SELECT → to display data
- WHERE → to filter records
- BETWEEN → to filter by date range
- ORDER BY → to sort data
- LIMIT → to get top results
- SUM → to perform calculations

ADVANCED SQL QUERY SUMMARY

- JOIN → to combine data from multiple tables
- GROUP BY → to group data
- HAVING → to filter grouped data
- Aggregate functions → like SUM, AVG



CREATE TABLE IN SQL



```sql
CREATE DATABASE OnlineBookstore;
CREATE TABLE Books (
    Book_ID SERIAL PRIMARY KEY,
    Title VARCHAR(100),
    Author VARCHAR(100),
    Genre VARCHAR(50),
    Published_Year INT,
    Price NUMERIC(10, 2),
    Stock INT
);
CREATE TABLE Customers (
    Customer_ID SERIAL PRIMARY KEY,
    Name VARCHAR(100),
    Email VARCHAR(100),
    Phone VARCHAR(15),
    City VARCHAR(50),
    Country VARCHAR(150)
);

CREATE TABLE Orders (
    Order_ID SERIAL PRIMARY KEY,
    Customer_ID INT REFERENCES Customers(Customer_ID),
    Book_ID INT REFERENCES Books(Book_ID),
    Order_Date DATE,
    Quantity INT,
    Total_Amount NUMERIC(10, 2)
);
```

SELECT * FROM BOOK;

Book_ID	Title	Author	Genre	Published_Year	Price	Stock
1	Configurable modular throughput	Joseph Crane	Biography	1949	21.34	100
2	Persevering reciprocal knowledge user	Mario Moore	Fantasy	1971	35.8	19
3	Streamlined coherent initiative	Derrick Howard	Non-Fiction	1913	15.75	27
4	Customizable 24hour product	Christopher Andrews	Fiction	2020	43.52	8
5	Adaptive 5thgeneration encoding	Juan Miller	Fantasy	1956	10.95	16
6	Advanced encompassing implementation	Bryan Morgan	Biography	1985	6.56	2
7	Open-architected exuding structure	Jacqueline Young	Romance	1927	43.63	95
8	Persistent local encoding	Troy Cox	Science Fiction	2019	48.99	84
9	Optimized interactive challenge	Colin Buckley	Fantasy	1987	14.33	70
10	Ergonomic national hub	Samantha Ruiz	Mystery	2015	24.63	25

SELECT * FROM CUSTOMERS;

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetc

Customer_ID	Name	Email	Phone	city	country
1	Deborah Griffith	balljoseph@wright-keith.net	1234567891	South Craigfort	Denmark
2	Crystal Clements	kimberlybennett@curtis.com	1234567892	East Derekberg	Nicaragua
3	Susan Fuller	beanmichael@burnett-stewart.net	1234567893	Austinbury	Equatorial Guinea
4	Jamie Ramirez	amandahood@warren.com	1234567894	Dianamouth	Slovenia
5	Marcus Murphy	connerjohn@yahoo.com	1234567895	Smithbury	Guinea-Bissau
6	Stephen Vasquez	ricemiguel@yahoo.com	1234567896	Hamiltonstad	Rwanda
7	Susan Hicks	jeffrey91@yahoo.com	1234567897	East Rebecca	Montenegro
8	Matthew Johnson	austinkenneth@manning.net	1234567898	Kirstenborough	Israel
9	Matthew Williams	jeffrey41@diaz.com	1234567899	Rebeccafurt	Somalia

SELECT * FROM ORDERS;

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Order_ID	Customer_ID	Book_ID	Order_Date	Quantity	Total_Amount
1	84	169	26-05-2023	8	188.56
2	137	301	23-01-2023	10	216.6
3	216	261	27-05-2024	6	85.5
4	433	343	25-11-2023	7	301.21
5	14	431	26-07-2023	7	136.36
6	439	119	11-10-2024	5	249.4
7	195	467	23-10-2023	6	82.92
8	32	159	07-05-2024	4	144.84
9	109	407	04-01-2024	9	379.71

RETERIVE ALL BOOKS IN THE "FICTION" GENURE:

```
SELECT * FROM BOOK

WHERE GENRE= "FICTION";
```

Book_ID	Title	Author	Genre	Published_Year	Price	Stock
4	Customizable 24hour product	Christopher Andrews	Fiction	2020	43.52	8
22	Multi-layered optimizing migration	Wesley Escobar	Fiction	1908	39.23	78
28	Expanded analyzing portal	Lisa Coffey	Fiction	1941	37.51	79
29	Quality-focused multi-tasking challenge	Katrina Underwood	Fiction	1905	31.12	100
31	Implemented encompassing conglomeration	Melissa Taylor	Fiction	2010	21.23	44
39	Optimized national process improvement	Megan Goodwin	Fiction	1978	10.99	42
40	Adaptive didactic interface	Natalie Gonzalez	Fiction	1923	25.97	94
47	Reverse-engineered directional conglomeration	John Christian	Fiction	2006	20.37	90
62	Re-contextualized real-time strategy	Nicole Lynch	Fiction	1953	26.34	23
63	Polarized heuristic database	Franklin Mack	Fiction	1989	22.38	56

FIND BOOKS PUBLISHED AFTER THE YEAR 1950

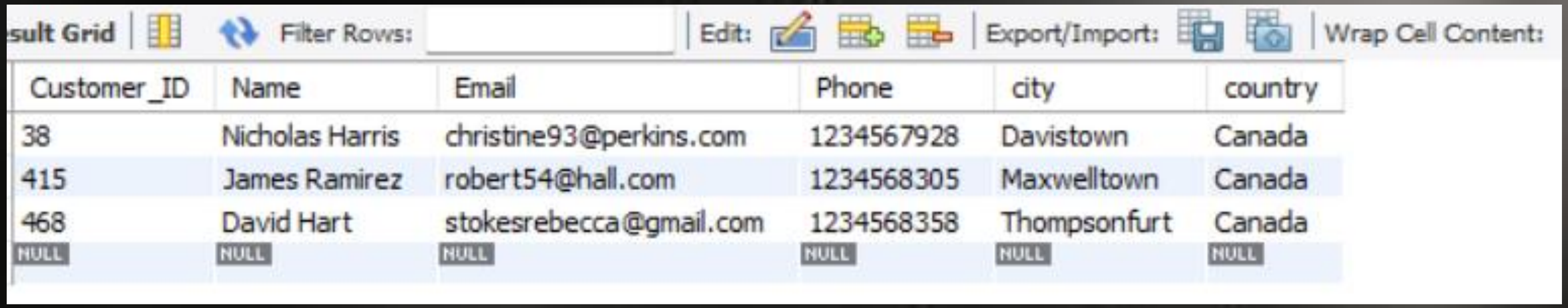
```
SELECT * FROM BOOK
WHERE PUBLISHED_YEAR >1950;
```

Book_ID	Title	Author	Genre	Published_Year	Price	Stoc
2	Persevering reciprocal knowledge user	Mario Moore	Fantasy	1971	35.8	19
4	Customizable 24hour product	Christopher Andrews	Fiction	2020	43.52	8
5	Adaptive 5thgeneration encoding	Juan Miller	Fantasy	1956	10.95	16
6	Advanced encompassing implementation	Bryan Morgan	Biography	1985	6.56	2
8	Persistent local encoding	Troy	Science Fiction	2019	48.99	84
9	Optimized interactive challenge	Colin Buckley	Fantasy	1987	14.33	70
10	Ergonomic national hub	Samantha Ruiz	Mystery	2015	24.63	25
11	Secured zero tolerance time-frame	Denise Barnes	Fantasy	1998	35.95	10
12	Polarized optimal array	Destiny Scott	Non-Fiction	1989	27.43	63

Troy Cox

LIST ALL CUSTOMERS FROM THE CANADA

```
SELECT * FROM CUSTOMERS
WHERE COUNTRY = "CANADA";
```



Customer_ID	Name	Email	Phone	city	country
38	Nicholas Harris	christine93@perkins.com	1234567928	Davistown	Canada
415	James Ramirez	robert54@hall.com	1234568305	Maxwelltown	Canada
468	David Hart	stokesrebecca@gmail.com	1234568358	Thompsonfurt	Canada
NULL	NULL	NULL	NULL	NULL	NULL

SHOW ORDER PLACED IN NOVEMBER 2023

```
SELECT * FROM ORDERS
WHERE ORDER_DATE BETWEEN '01-11-2023'AND '30-11-2023';
```

Order_ID	Customer_ID	Book_ID	Order_Date	Quantity	Total_Amount
1	84	169	26-05-2023	8	188.56
2	137	301	23-01-2023	10	216.6
3	216	261	27-05-2024	6	85.5
4	433	343	25-11-2023	7	301.21
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7	195	467	23-10-2023	6	82.92
8	32	159	07-05-2024	4	144.84
9	109	407	04-01-2024	9	379.71

RETERIVE THE TOTAL STOCK OF BOOKS AVAILABLE;

```
SELECT SUM(STOCK) AS TOTAL_STOCK
FROM BOOK;
```



total_stock
25056

SHOW ALL CUSTOMERS WHO ORDERED MORE THEN ONE QUANTITY OF A BOOK

```
SELECT * FROM ORDERS
WHERE QUANTITY >I;
```

Order_ID	Customer_ID	Book_ID	Order_Date	Quantity	Total_Amount
1	84	169	26-05-2023	8	188.56
2	137	301	23-01-2023	10	216.6
3	216	261	27-05-2024	6	85.5
4	433	343	25-11-2023	7	301.21
5	14	431	26-07-2023	7	136.36
6	439	119	11-10-2024	5	249.4
7	195	467	23-10-2023	6	82.92
8	32	159	07-05-2024	4	144.84
9	109	407	04-01-2024	9	379.71
10	94	122	09-07-2024	4	123

RETERIVE ALL THE ORDERS WHERE TOTAL AMOUNT EXCEDDS $20

```
SELECT * FROM ORDERS
WHERE TOTAL_AMOUNT>20;
```

	Order_ID	Customer_ID	Book_ID	Order_Date	Quantity	Total_Amount
▶	1	84	169	26-05-2023	8	188.56
	2	137	301	23-01-2023	10	216.6
	3	216	261	27-05-2024	6	85.5
	4	433	343	25-11-2023	7	301.21
	5	14	431	26-07-2023	7	136.36
	6	439	119	11-10-2024	5	249.4
	7	195	467	23-10-2023	6	82.92
	8	32	159	07-05-2024	4	144.84
	9	109	407	04-01-2024	9	379.71

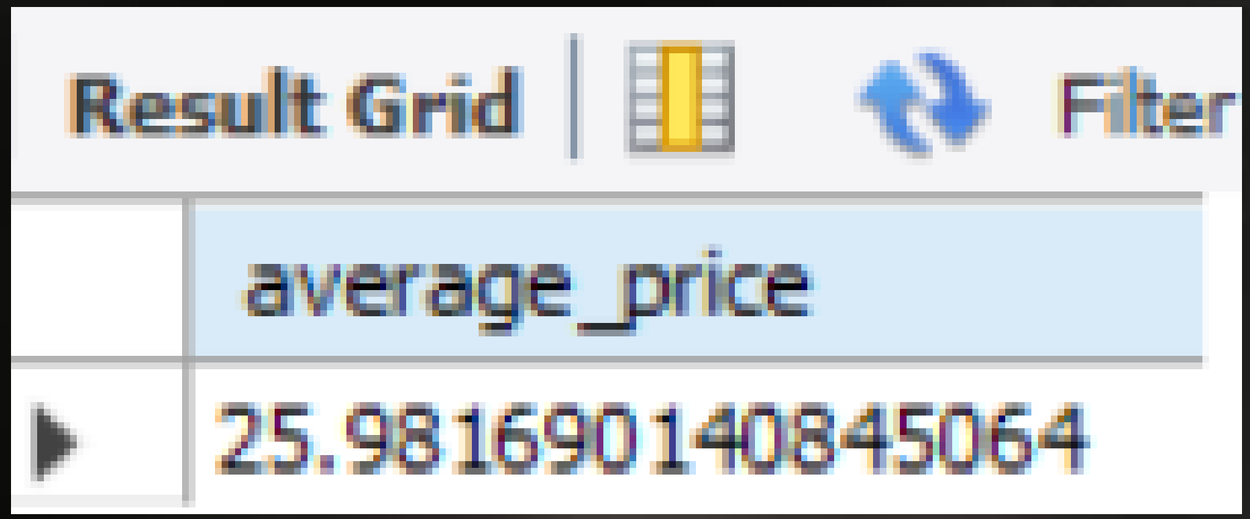
RETERIVE THE TOTAL NUMBER OF BOOKS SOLD FOR EACH GENRE

```
SELECT B.GENRE, SUM(O.QUANTITY) AS TOTAL_BOOK_SOLD
FROM ORDERS O
JOIN BOOK B ON O.BOOK_ID = B.BOOK_ID
GROUP BY B.GENRE;
```

genre	total_book_sold
Biography	279
Non-Fiction	351
Fantasy	446
Romance	439
Science Fiction	447
Mystery	504
Fiction	225

FIND THE AVERAGE PRICE OF BOOKS IN THE "FANTASY" GENRE

```
SELECT  AVG(PRICE) AS AVERAGE_PRICE
FROM BOOK
WHERE GENRE = "FANTASY";
```



LIST CUSTOMERS WHO HAVE PLACED AT LEAST 2 ORDERS

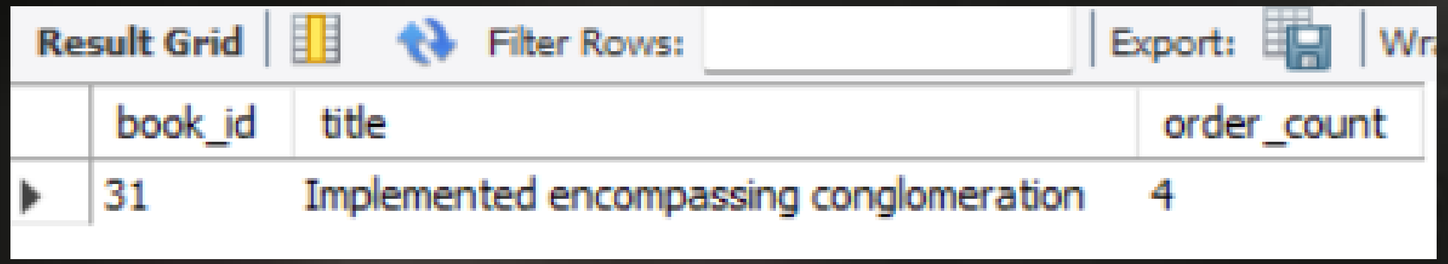
```
SELECT O.CUSTOMER_ID ,C.NAME,
COUNT(O.ORDER_ID) AS ORDER_COUNT
FROM ORDERS O
JOIN CUSTOMERS C ON
O.CUSTOMER_ID=C.CUSTOMER_ID
GROUP BY O.CUSTOMER_ID,C.NAME
HAVING COUNT(ORDER_ID) >=2;
```

customer_id	name	order_count
84	Gary Blair	2
137	Steven Miller	2
216	Phillip Allen	2
14	John Wood	2
195	Dominique Turner	3
109	Jacob Kelley	2
94	Mr. David Cox	3
131	Peter Smith	2
454	April Anderson	2
420	Andrew Murrav	3

sult 35

FIND MOST FREQUENTLY ORDER BOOK

```
SELECT
O.BOOK_ID,B.TITLE,COUNT(O.ORDER_ID)
AS ORDER_COUNT
FROM ORDERS O
JOIN BOOK B ON O.BOOK_ID = B.BOOK_ID
GROUP BY O.BOOK_ID,B.TITLE
ORDER BY ORDER_COUNT DESC LIMIT I;
```



SHOW THE TOP 3 MOST EXPENSIVE BOOK SOLD BY EACH AUTHOR;

```
SELECT * FROM BOOK
WHERE GENRE = "FANTASY"
ORDER BY PRICE DESC LIMIT 3 ;
```

Book_ID	Title	Author	Genre	Published_Year	Price	Stock
240	Stand-alone content-based hub	Lisa Ellis	Fantasy	1957	49.9	41
462	Innovative 3rdgeneration database	Allison Contreras	Fantasy	1988	49.23	62
238	Optimized even-keeled analyzer	Sherri Griffith	Fantasy	1975	48.97	72

RETERIVE THE TOTAL QUANTITY OF BOOK SOLD BY EACH AUTHOR

```
SELECT B.AUTHOR, SUM(O.QUANTITY) AS
TOTAL_BOOK_SOLD
FROM ORDERS O
JOIN BOOK B  ON O.BOOK_ID=B.BOOK_ID
GROUP BY B.AUTHOR;
```

author	total_book_sold
Joseph Crane	3
Derrick Howard	5
Juan Miller	8
Jacqueline Young	5
Troy Cox	3
Samantha Ruiz	1
Denise Barnes	5
Jaclyn Miller	9
Christopher Price	1
Beniamin Peters	9

Result 38

LIST THE CITIES WHERE CUSTOMERS WHO SPENT OVER $30 ARE LOCATED

```
SELECT DISTINCT C.CITY,TOTAL_AMOUNT
FROM ORDERS O
JOIN CUSTOMERS C ON
O.CUSTOMER_ID=C.CUSTOMER_ID
WHERE O.TOTAL_AMOUNT >30;
```

city	total_amou
Lake Paul	188.56
North Keith	216.6
Kelseyfort	85.5
East David	301.21
Richardsonville	136.36
Ramosstad	249.4
Rogersborough	82.92
New Carlosbury	144.84
Ravenberg	379.71
West Anthony	123

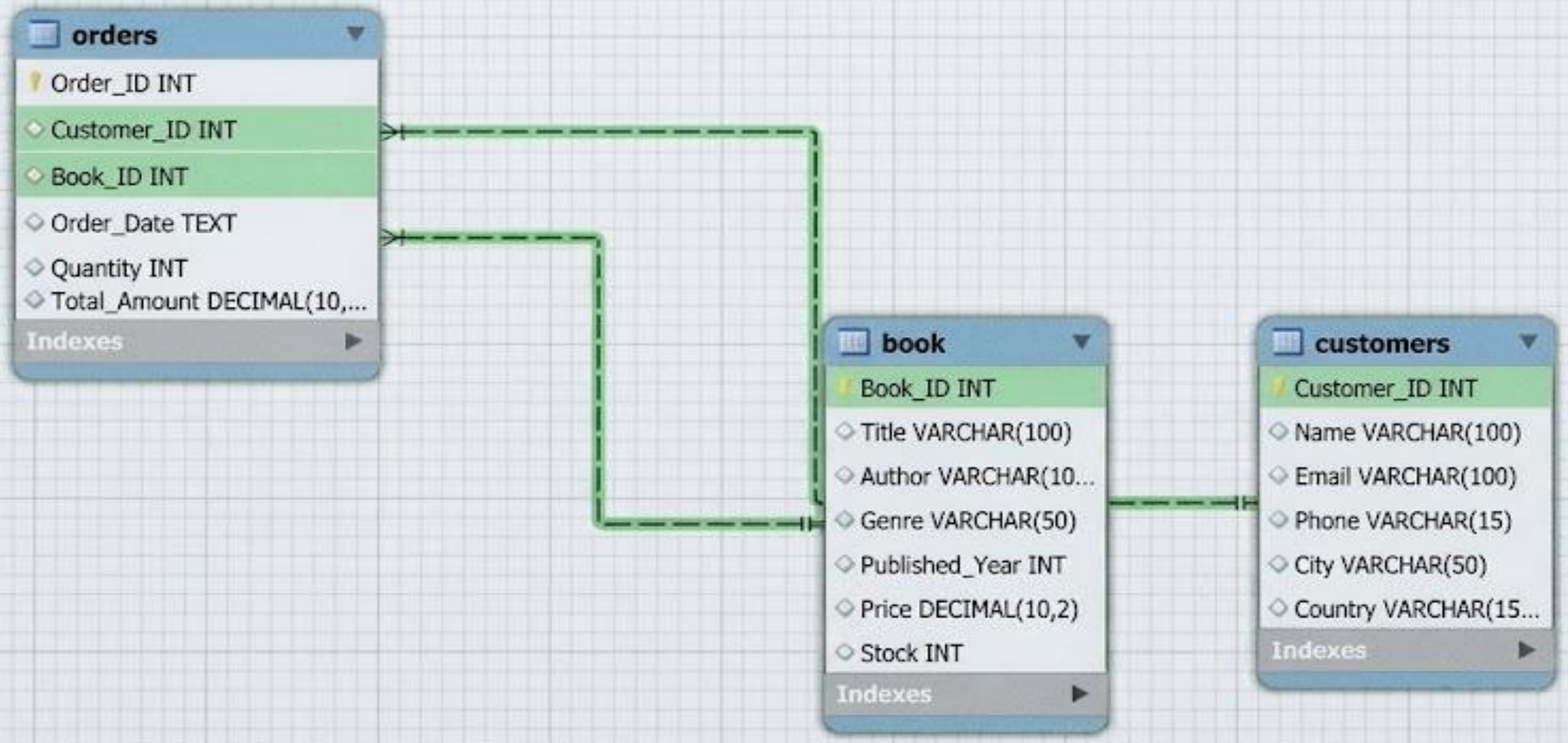
FIND THE CUSTOMERS WHO SPENT THE MOST ON ORDERS

```
SELECT C.CUSTOMER_ID,C.NAME,SUM(O.TOTAL_AMOUNT) AS TOTAL_SPENT
FROM ORDERS O
JOIN CUSTOMERS C ON O.CUSTOMER_ID = C.CUSTOMER_ID
GROUP BY C.CUSTOMER_ID, C.NAME
ORDER BY TOTAL_SPENT DESC LIMIT I;
```

customer_id	name	total_spent
457	Kim Turner	1398.9

Entity-Relationship (ER) Diagram: Online Bookstore

Database Schema for This Project




Thank You